Bayes' Theorem

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Lecture objectives

By the end of this lecture, you should be able to:

=> Explain conditional probability and the multiplication rule.

=> State and prove Bayes' Theorem.

=> Use the law of total probability to compute evidence probabilities.

=> Solve Bayes' Theorem problems involving tests, choices, boxes, cards and real-life decisions.

=> Distinguish between prior probability, likelihood and posterior probability.

Motivating question

Question

A person tests positive for a rare disease. The test is fairly accurate.

- Does a positive result mean the person almost certainly has the disease?
- What other information do we need?

$\Rightarrow$ Bayes' Theorem helps us update probability when new evidence is observed.

$\Rightarrow$ It combines the original chance of an event with the reliability of the evidence.

Key vocabulary

Main terms

=> **Prior probability**: what we believe before seeing evidence, for example $P(A)$.

=> **Evidence**: the observed event, for example B.

=> **Likelihood**: probability of the evidence if the hypothesis is true, $P(B \mid A)$.

=> **Posterior probability**: updated probability after seeing evidence, $P(A \mid B)$.

Conditional probability

Definition

The conditional probability of event A given that event B has occurred is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

$\Rightarrow$ $P(A \mid B)$ means the probability of A after restricting attention to outcomes in B.

$\Rightarrow$ The denominator $P(B)$ must be positive.

Multiplication rule

From the definition of conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Therefore,

$$P(A \cap B) = P(A \mid B)P(B).$$

Similarly,

$$P(A \cap B) = P(B \mid A)P(A).$$

Important idea

The same intersection $A \cap B$ can be written in two ways.

Example: conditional probability

Example

A card is drawn from a standard deck of 52 cards. Let A be the event that the card is a king, and B the event that the card is a face card. Find $P(A \mid B)$.

Solution

There are 12 face cards: J, Q, K in four suits. There are 4 kings.

$$P(A \mid B) = \frac{4}{12} = \frac{1}{3}.$$

Bayes' Theorem

Theorem

For events A and B such that $P(A) > 0$ and $P(B) > 0$,

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

$\Rightarrow$ Bayes' Theorem reverses the condition.

$\Rightarrow$ It allows us to compute $P(A \mid B)$ using $P(B \mid A)$.

Proof of Bayes' Theorem

$\Rightarrow$ By conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

$\Rightarrow$ Also,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

$\Rightarrow$ Hence,

$$P(A \cap B) = P(B \mid A)P(A).$$

$\Rightarrow$ Substituting gives

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

Law of total probability

Suppose $A_1, A_2, \ldots, A_n$ are mutually exclusive and exhaustive events.

Total probability

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i) P(A_i).$$

$=>$ Mutually exclusive: no two A_i events occur at the same time.

$=>$ Exhaustive: one of the A_i events must occur.

Generalized Bayes' Theorem

If $A_1, A_2, \ldots, A_n$ are mutually exclusive and exhaustive, then for $k = 1, 2, \ldots, n$,

$$P(A_k \mid B) = \frac{P(B \mid A_k)P(A_k)}{\sum_{i=1}^{n} P(B \mid A_i)P(A_i)}.$$

Evidence term

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)P(A_i).$$

How to solve Bayes' Theorem problems

1. Define the events clearly.
2. Identify the prior probabilities, such as $P(A)$ and $P(A^c)$.
3. Identify the likelihoods, such as $P(B \mid A)$ and $P(B \mid A^c)$.
4. Compute the total probability of the evidence $P(B)$.
5. Substitute into Bayes' formula.
6. Interpret the final answer in words.

Bayes example template: formula first

For every Bayes' Theorem example, use this structure:

1. Define the hypotheses $H_1, H_2, \ldots, H_n$.
2. Identify the **prior**: $P(H_i)$.
3. Identify the **evidence**: E.
4. Identify the **posterior**: $P(H_k \mid E)$.
5. Write total probability before substitution:

$$P(E) = \sum_{i=1}^{n} P(E \mid H_i) P(H_i).$$

6. Write Bayes' formula with the denominator expanded, then punch in numbers.

Example 1: two buckets

Example

There are two buckets.

- Bucket B_1 contains 3 red balls and 7 blue balls.
- Bucket B_2 contains 6 red balls and 4 blue balls.

A bucket is chosen at random, and a red ball is drawn. Find $P(B_2 \mid R)$.

Example 1: identify the Bayes parts

=> **Hypotheses:** B_1 = bucket 1 was chosen, B_2 = bucket 2 was chosen.

=> **Evidence:** R = a red ball was drawn.

=> **Prior probabilities:**

$$P(B_1) = \frac{1}{2}, \qquad P(B_2) = \frac{1}{2}.$$

=> **Posterior probability required:**

$$P(B_2 \mid R).$$

=> **Likelihoods:**

$$P(R \mid B_1) = \frac{3}{10}, \qquad P(R \mid B_2) = \frac{6}{10}.$$

Example 1: formula before substitution

Prior: $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{2}$. **Evidence:** R. **Posterior:** $P(B_2 \mid R)$.
Total probability of the evidence before substitution:

$$P(R) = P(R \mid B_1)P(B_1) + P(R \mid B_2)P(B_2).$$

Bayes formula with total probability in the denominator:

$$P(B_2 \mid R) = \frac{P(R \mid B_2)P(B_2)}{P(R \mid B_1)P(B_1) + P(R \mid B_2)P(B_2)}.$$

Punch in the numbers:

$$P(B_2 \mid R) = \frac{\frac{6}{10} \times \frac{1}{2}}{\frac{3}{10} \times \frac{1}{2} + \frac{6}{10} \times \frac{1}{2}} = \frac{\frac{6}{20}}{\frac{9}{20}} = \frac{2}{3} \approx 0.6667.$$

Example 1: interpretation

Conclusion

Given that the ball drawn is red, the probability that it came from bucket B_2 is

$$\boxed{\frac{2}{3} \approx 66.67\%}.$$

$=>$ Before drawing, bucket B_2 had probability 50%.

$=>$ After observing a red ball, the probability rises to 66.67% because bucket B_2 has more red balls.

Example 2: cancer test

Example

A certain cancer has prevalence 1% in the population. A diagnostic test has:

- true positive rate: $P(+ \mid C) = 0.90$;
- true negative rate: $P(- \mid C^c) = 0.95$.

If a randomly selected person tests positive, find $P(C \mid +)$.

Example 2: identify the Bayes parts

$\Rightarrow$ **Hypotheses:** C = person has cancer, C^c = person does not have cancer.

$\Rightarrow$ **Evidence:** $+$ = the test result is positive.

$\Rightarrow$ **Prior probabilities:**

$$P(C) = 0.01, \qquad P(C^c) = 0.99.$$

$\Rightarrow$ **Posterior probability required:**

$$P(C \mid +).$$

$\Rightarrow$ **Likelihoods:**

$$P(+ \mid C) = 0.90, \qquad P(+ \mid C^c) = 1 - 0.95 = 0.05.$$

Example 2: formula before substitution

Prior: $P(C) = 0.01$, $P(C^c) = 0.99$. **Evidence:** $+$. **Posterior:** $P(C \mid +)$.
Total probability of the evidence before substitution:

$$P(+) = P(+ \mid C)P(C) + P(+ \mid C^c)P(C^c).$$

Bayes formula with total probability in the denominator:

$$P(C \mid +) = \frac{P(+ \mid C)P(C)}{P(+ \mid C)P(C) + P(+ \mid C^c)P(C^c)}.$$

Punch in the numbers:

$$P(C \mid +) = \frac{0.90(0.01)}{0.90(0.01) + 0.05(0.99)} = \frac{0.009}{0.0585} \approx 0.1538.$$

Example 2: interpretation

Conclusion

$$\boxed{P(C \mid +) \approx 15.38\%}$$

$\Rightarrow$ The prior probability was only 1% because the disease is rare.

$\Rightarrow$ A positive test result increases the probability, but not to 90%.

$\Rightarrow$ The false-positive term contributes strongly to the evidence probability $P(+)$.

Example 3: machine production

Example

A factory has two machines. Machine M_1 produces 70% of items and machine M_2 produces 30%.

- M_1 has defect rate 2%.
- M_2 has defect rate 5%.

An item is selected and found to be defective. Find $P(M_2 \mid D)$.

Example 3: identify the Bayes parts

=> **Hypotheses:** M_1 = item made by machine 1, M_2 = item made by machine 2.

=> **Evidence:** D = selected item is defective.

=> **Prior probabilities:**

$$P(M_1) = 0.70, \qquad P(M_2) = 0.30.$$

=> **Posterior probability required:**

$$P(M_2 \mid D).$$

=> **Likelihoods:**

$$P(D \mid M_1) = 0.02, \qquad P(D \mid M_2) = 0.05.$$

Example 3: formula before substitution

Prior: $P(M_1) = 0.70$, $P(M_2) = 0.30$. **Evidence:** D. **Posterior:** $P(M_2 \mid D)$.
Total probability of the evidence before substitution:

$$P(D) = P(D \mid M_1)P(M_1) + P(D \mid M_2)P(M_2).$$

Bayes formula with total probability in the denominator:

$$P(M_2 \mid D) = \frac{P(D \mid M_2)P(M_2)}{P(D \mid M_1)P(M_1) + P(D \mid M_2)P(M_2)}.$$

Punch in the numbers:

$$P(M_2 \mid D) = \frac{0.05(0.30)}{0.02(0.70) + 0.05(0.30)} = \frac{0.015}{0.029} \approx 0.5172.$$

Example 3: interpretation

Conclusion

$$\boxed{P(M_2 \mid D) \approx 51.72\%}$$

$=>$ Machine M_2 produces fewer items, but has a higher defect rate.

$=>$ Once a defective item is observed, machine M_2 becomes slightly more likely to be the source.

Example 4: email spam filter

Example

Suppose 30% of emails are spam. A spam filter marks:

- 95% of spam emails as spam;
- 10% of non-spam emails as spam.

If an email is marked as spam, find the probability that it is actually spam.

Example 4: identify the Bayes parts

Let S be the event that an email is spam and M the event that it is marked as spam.

$\Rightarrow$ **Hypotheses:** S = spam, S^c = not spam.

$\Rightarrow$ **Evidence:** M = email is marked as spam.

$\Rightarrow$ **Prior probabilities:**

$$P(S) = 0.30, \qquad P(S^c) = 0.70.$$

$\Rightarrow$ **Posterior probability required:**

$$P(S \mid M).$$

$\Rightarrow$ **Likelihoods:**

$$P(M \mid S) = 0.95, \qquad P(M \mid S^c) = 0.10.$$

Example 4: formula before substitution

Prior: $P(S) = 0.30$, $P(S^c) = 0.70$. **Evidence:** M. **Posterior:** $P(S \mid M)$.
Total probability of the evidence before substitution:

$$P(M) = P(M \mid S)P(S) + P(M \mid S^c)P(S^c).$$

Bayes formula with total probability in the denominator:

$$P(S \mid M) = \frac{P(M \mid S)P(S)}{P(M \mid S)P(S) + P(M \mid S^c)P(S^c)}.$$

Punch in the numbers:

$$P(S \mid M) = \frac{0.95(0.30)}{0.95(0.30) + 0.10(0.70)} = \frac{0.285}{0.355} \approx 0.8028.$$

Example 4: interpretation

Conclusion

$$\boxed{P(S \mid M) \approx 80.28\%}$$

$\Rightarrow$ Marking an email as spam is strong evidence, but not perfect evidence.

$\Rightarrow$ False positives from non-spam emails reduce the posterior probability.

Example 5: dice and hidden choice

Example

A bag contains two dice:

- Die D_1 is fair.
- Die D_2 is loaded so that $P(6 \mid D_2) = \frac{1}{2}$.

A die is chosen at random and rolled once. The result is 6. Find $P(D_2 \mid 6)$.

Example 5: identify the Bayes parts

=> **Hypotheses:** D_1 = fair die was chosen, D_2 = loaded die was chosen.

=> **Evidence:** E = a 6 is rolled.

=> **Prior probabilities:**

$$P(D_1) = \frac{1}{2}, \qquad P(D_2) = \frac{1}{2}.$$

=> **Posterior probability required:**

$$P(D_2 \mid E).$$

=> **Likelihoods:**

$$P(E \mid D_1) = \frac{1}{6}, \qquad P(E \mid D_2) = \frac{1}{2}.$$

Example 5: formula before substitution

Prior: $P(D_1) = \frac{1}{2}$, $P(D_2) = \frac{1}{2}$. **Evidence:** $E =$ a 6 is rolled. **Posterior:** $P(D_2 \mid E)$.
Total probability of the evidence before substitution:

$$P(E) = P(E \mid D_1)P(D_1) + P(E \mid D_2)P(D_2).$$

Bayes formula with total probability in the denominator:

$$P(D_2 \mid E) = \frac{P(E \mid D_2)P(D_2)}{P(E \mid D_1)P(D_1) + P(E \mid D_2)P(D_2)}.$$

Punch in the numbers:

$$P(D_2 \mid E) = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{1}{6} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{1}{3}} = \frac{3}{4}.$$

Example 5: interpretation

Conclusion

$$\boxed{P(D_2 \mid E) = \frac{3}{4} = 75\%}$$

$=>$ Before rolling, both dice were equally likely.

$=>$ Rolling a 6 gives stronger support to the loaded die.

Example 6: cards

Example

A box contains two decks:

- Deck A is a standard 52-card deck.
- Deck B is a 52-card deck with all hearts replaced by spades.

A deck is selected at random and one card is drawn. The card is a spade. Find $P(B \mid S)$.

Example 6: identify the Bayes parts

=> **Hypotheses:** A = standard deck chosen, B = modified deck chosen.

=> **Evidence:** S = the card drawn is a spade.

=> **Prior probabilities:**

$$P(A) = \frac{1}{2}, \qquad P(B) = \frac{1}{2}.$$

=> **Posterior probability required:**

$$P(B \mid S).$$

=> **Likelihoods:**

$$P(S \mid A) = \frac{13}{52} = \frac{1}{4}, \qquad P(S \mid B) = \frac{26}{52} = \frac{1}{2}.$$

Example 6: formula before substitution

Prior: $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$. **Evidence:** $S =$ the card is a spade. **Posterior:** $P(B \mid S)$.

Total probability of the evidence before substitution:

$$P(S) = P(S \mid A)P(A) + P(S \mid B)P(B).$$

Bayes formula with total probability in the denominator:

$$P(B \mid S) = \frac{P(S \mid B)P(B)}{P(S \mid A)P(A) + P(S \mid B)P(B)}.$$

Punch in the numbers:

$$P(B \mid S) = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{1}{4} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{3}{8}} = \frac{2}{3}.$$

Example 6: interpretation

Conclusion

$$\boxed{P(B \mid S) = \frac{2}{3} \approx 66.67\%}$$

$\Rightarrow$ The modified deck has twice as many spades as the standard deck.

$\Rightarrow$ Observing a spade therefore increases the probability that deck B was chosen.

Independence of events

Definition

Two events A and B are independent if

$$P(A \mid B) = P(A)$$

and

$$P(B \mid A) = P(B).$$

$=>$ If two events are independent, knowing one occurred does not change the probability of the other.

Example: two coin tosses

Toss a fair coin twice. Define:

$$A = \{\text{first toss is Head}\}, \quad B = \{\text{second toss is Head}\}.$$

Sample space:

$$S = \{HH, HT, TH, TT\}.$$

$$P(A) = \frac{2}{4} = \frac{1}{2}, \quad P(B) = \frac{2}{4} = \frac{1}{2}.$$

$$A \cap B = \{HH\}, \quad P(A \cap B) = \frac{1}{4}.$$

Example: proof of independence

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} = P(A).$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} = P(B).$$

Conclusion

The event that the first toss is Head is independent of the event that the second toss is Head.

Bayes' Theorem when events are independent

If A and B are independent, then $P(B \mid A) = P(B)$.
Bayes' Theorem becomes:

$$P(A \mid B) = \frac{P(B)P(A)}{P(B)} = P(A).$$

Meaning

If the evidence B is independent of the event A, then observing B does not update the probability of A.

Practice question 1: bags of balls

Bag A contains 4 red and 6 blue balls. Bag B contains 8 red and 2 blue balls. A bag is selected at random and a red ball is drawn.

Question

Find the probability that the selected bag was Bag B.

Practice 1: formula before substitution

Prior: $P(A) = P(B) = \frac{1}{2}$. **Evidence:** $R =$ red ball drawn. **Posterior:** $P(B \mid R)$.
Total probability before substitution:

$$P(R) = P(R \mid A)P(A) + P(R \mid B)P(B).$$

Bayes formula before substitution:

$$P(B \mid R) = \frac{P(R \mid B)P(B)}{P(R \mid A)P(A) + P(R \mid B)P(B)}.$$

Punch in the numbers:

$$P(B \mid R) = \frac{\frac{8}{10} \times \frac{1}{2}}{\frac{4}{10} \times \frac{1}{2} + \frac{8}{10} \times \frac{1}{2}} = \frac{2}{3}.$$

Practice question 2: defective laptop

A shop receives laptops from Supplier A and Supplier B.

- Supplier A provides 60% of laptops and has defect rate 3%.
- Supplier B provides 40% of laptops and has defect rate 6%.

A laptop is defective. Find the probability that it came from Supplier B.

Practice 2: formula before substitution

Prior: $P(A) = 0.60$, $P(B) = 0.40$. **Evidence:** $D =$ defective laptop. **Posterior:** $P(B \mid D)$.

Total probability before substitution:

$$P(D) = P(D \mid A)P(A) + P(D \mid B)P(B).$$

Bayes formula before substitution:

$$P(B \mid D) = \frac{P(D \mid B)P(B)}{P(D \mid A)P(A) + P(D \mid B)P(B)}.$$

Punch in the numbers:

$$P(B \mid D) = \frac{0.06(0.40)}{0.03(0.60) + 0.06(0.40)} = \frac{0.024}{0.042} = \frac{4}{7}.$$

Practice question 3: exam preparation

A student either studies well or does not study well.

- $P(S) = 0.65$.
- If the student studies well, $P(\text{pass} \mid S) = 0.90$.
- If the student does not study well, $P(\text{pass} \mid S^c) = 0.40$.

If the student passes, find $P(S \mid \text{pass})$.

Practice 3: formula before substitution

Let P be the event that the student passes. **Prior:** $P(S) = 0.65$, $P(S^c) = 0.35$.
Evidence: P. **Posterior:** $P(S \mid P)$.
Total probability before substitution:

$$P(P) = P(P \mid S)P(S) + P(P \mid S^c)P(S^c).$$

Bayes formula before substitution:

$$P(S \mid P) = \frac{P(P \mid S)P(S)}{P(P \mid S)P(S) + P(P \mid S^c)P(S^c)}.$$

Punch in the numbers:

$$P(S \mid P) = \frac{0.90(0.65)}{0.90(0.65) + 0.40(0.35)} \approx 0.8069.$$

Practice question 4: three boxes

There are three boxes:

- Box 1: 2 red and 8 blue balls.
- Box 2: 5 red and 5 blue balls.
- Box 3: 9 red and 1 blue ball.

A box is selected at random and a red ball is drawn.

Question

Find $P(\text{Box } 3 \mid R)$.

Practice 4: formula before substitution

Prior: $P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$. **Evidence:** R. **Posterior:** $P(B_3 \mid R)$.
Total probability before substitution:

$$P(R) = P(R \mid B_1)P(B_1) + P(R \mid B_2)P(B_2) + P(R \mid B_3)P(B_3).$$

Bayes formula before substitution:

$$P(B_3 \mid R) = \frac{P(R \mid B_3)P(B_3)}{P(R \mid B_1)P(B_1) + P(R \mid B_2)P(B_2) + P(R \mid B_3)P(B_3)}.$$

Punch in the numbers:

$$P(B_3 \mid R) = \frac{\frac{9}{10} \times \frac{1}{3}}{\frac{2}{10} \times \frac{1}{3} + \frac{5}{10} \times \frac{1}{3} + \frac{9}{10} \times \frac{1}{3}} = \frac{9}{16}.$$

Practice question 5: cybersecurity alert

A network intrusion detection system is used to detect attacks.

- 2% of network sessions are attacks.
- The system detects 98% of attacks.
- It incorrectly flags 4% of normal sessions as attacks.

Question

If the system raises an alert, what is the probability that the session is actually an attack?

Practice 5: formula before substitution

Let A be an attack and F be a flag. **Prior:** $P(A) = 0.02$, $P(A^c) = 0.98$. **Evidence:** F. **Posterior:** $P(A \mid F)$.

Total probability before substitution:

$$P(F) = P(F \mid A)P(A) + P(F \mid A^c)P(A^c).$$

Bayes formula before substitution:

$$P(A \mid F) = \frac{P(F \mid A)P(A)}{P(F \mid A)P(A) + P(F \mid A^c)P(A^c)}.$$

Punch in the numbers:

$$P(A \mid F) = \frac{0.98(0.02)}{0.98(0.02) + 0.04(0.98)} = \frac{1}{3}.$$

Common mistakes

$\Rightarrow$ Confusing $P(A \mid B)$ with $P(B \mid A)$.

$\Rightarrow$ Forgetting to compute $P(B)$ using total probability.

$\Rightarrow$ Ignoring the base rate or prior probability $P(A)$.

$\Rightarrow$ Treating a positive test result as proof of the condition.

$\Rightarrow$ Forgetting to define the events before substituting values.

Key formulas

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

Total probability

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)P(A_i).$$

Summary

$\Rightarrow$ Bayes' Theorem updates prior probability using new evidence.

$\Rightarrow$ The denominator is the total probability of observing the evidence.

$\Rightarrow$ It is useful in medical testing, spam filtering, cybersecurity, quality control and decision making.

$\Rightarrow$ Always interpret the numerical answer in the context of the problem.

Questions?